Now the listener moves directly toward the stationary source (Figure 11.45). If v_L is the speed of the listener, then the relative speed of sound with respect to the listener becomes $v' = v + v_L$. Since the wavelength remains unchanged (because the source is stationary), the frequency of sound observed by the listener is changed and the observed frequency f' is given by

$$f' = \frac{v'}{\lambda} = \frac{v + v_L}{\lambda}$$

Using the equation (11.83),

$$f' = \left(\frac{v + v_L}{v}\right) f \qquad (11.84)$$

(listener moving toward the source)

Thus, the observed frequency is greater than the source frequency when the listener moves toward the stationary source.

If the listener is moving away from the stationary source, the observed frequency can be obtained from equation (11.84) by taking negative value for v_L. It is given by

$$f' = \left(\frac{v + (-v_L)}{v}\right) f$$

$$f' = \left(\frac{v - v_L}{v}\right) f \qquad (11.85)$$

(listener moving away from the source)

Thus, the observed frequency is less than the source frequency when the listener is moving away from the stationary source.

ii) Observed frequency: Moving source and stationary listener

Assume that both the source S and the listener L are at rest as shown in Figure 11.46a. Two successive compressions are also shown and are represented by two concentric circles. The second compression has just been emitted and is still near the source. The distance between two successive compressions is the wavelength λ of the sound. Since f is the frequency of the source, then the time between emissions of compressions is

$$T = \frac{1}{f} = \frac{\lambda}{v}$$

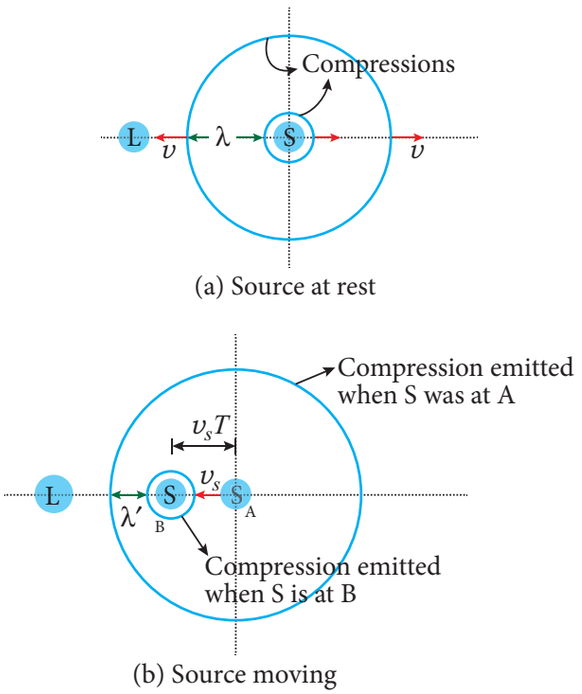


Figure 11.46 Source moves toward the stationary listener

Now the listener is stationary and the source moves directly toward the listener (Figure 11.46(b)). Let the speed of the source be v_S which is less than the speed of sound v.

In a time T, the first compression travels a distance $vT = \lambda$ and the source moves a distance $v_S T$. As a result, the distance between two successive compressions is decreased from λ to $\lambda' = \lambda - v_S T$. Therefore, the wavelength observed by the listener is given by

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